



- Used to avoid a shift or more than 90° as well as to avoid a double bit change which causes the output signal to shift by 180°
- ADVANTAGE:** Limited phase shift that is imparted during modulation
- DISADVANTAGE:** Requires a bandwidth twice that of conventional QPSK

8-PSK

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8-PSK is an M-ary encoding technique where M = 8. With an 8-PSK modulator, there are eight possible output phases. To encode eight different phases, the incoming bits are considered in groups of 3 bits called *tribits*.

Minimum Bandwidth Required (f_N)

$$f_N = 2 \left(\frac{f_b}{6} \right) = \frac{f_b}{3}$$

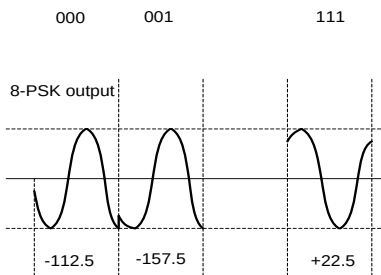
Output Baud Rate

$$\text{Output baud rate} = f_N = \frac{f_b}{3}$$

M-ary Encoding (For 8-PSK, M=8)

$$N = \log_2 M = \log_2 8 = 3 \text{ bits}$$

8-PSK Waveform



Binary Input			Output Phase
Q	I	C	
0	0	0	-112.5°
0	0	1	-157.5°
0	1	0	-67.5°
0	1	1	-22.5°
1	0	0	112.5°
1	0	1	157.5°
1	1	0	67.5°
1	1	1	22.5°

16-PSK

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16-PSK is an M-ary encoding technique where M = 16. A 16-PSK modulator acts on the incoming data in groups of 4 bits called *quadbits*, that is, for every four bits at the input there is one output produced.

Bandwidth Required (f_N)

$$f_N = 2 \left(\frac{f_b}{8} \right) = \frac{f_b}{4}$$

Output Baud Rate

$$\text{Output baud rate} = f_N = \frac{f_b}{4}$$

M-ary Encoding (For 16-PSK, M = 16)

$$N = \log_2 M = \log_2 16 = 4 \text{ bits}$$

Binary Input	Output Phase	Binary Input	Output Phase
0000	11.25°	1000	191.25°
0001	33.75°	1001	213.75°
0010	56.25°	1010	236.25°
0011	78.75°	1011	258.75°
0100	101.25°	1100	281.25°
0101	123.75°	1101	303.75°
0110	146.25°	1110	326.25°
0111	168.75°	1111	348.75°

Differential Binary PSK (DBPSK)

DBPSK is an alternative form of digital modulation where the binary input information is contained in the difference between two successive signaling elements rather than the absolute phase. The difference in the phase of the two signaling elements determines the logic condition of the data.

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Quadrature Amplitude Modulation (QAM)

QAM is a form of digital modulation where the digital information is contained in both the amplitude and phase of the transmitted carrier hence a combination of ASK and PSK techniques. It is considered a hybrid digital modulation scheme since it combines ASK and PSK methods.

8-QAM

8-QAM is an M-ary encoding technique where M = 8. Unlike 8-PSK, the output of an 8-QAM modulator is not a constant-amplitude signal.

Minimum Bandwidth Required (f_N)

$$f_N = 2 \left(\frac{f_b}{6} \right) = \frac{f_b}{3}$$

Output Baud Rate

$$\text{Output baud rate} = f_N = \frac{f_b}{3}$$

M-ary Encoding (For 8-QAM, M = 8)

$$N = \log_2 M = \log_2 8 = 3 \text{ bits}$$

Binary Input			Output Phase	Output Amplitude
Q	I	C		
0	0	0	-135°	0.765 V
0	0	1	-135°	1.848 V
0	1	0	-45°	0.765 V
0	1	1	-45°	1.848 V
1	0	0	135°	0.765 V
1	0	1	135°	1.848 V
1	1	0	45°	0.765 V
1	1	1	45°	1.848 V

16-QAM

16-QAM is an M-ary encoding technique where M = 16. The input data are acted by the modulator in groups of four bits called *quadbits* similar to 16-PSK.

Minimum Bandwidth Required (f_N)

$$f_N = 2 \left(\frac{f_b}{8} \right) = \frac{f_b}{4}$$

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Output Baud Rate

$$\text{Output baud rate} = f_N = \frac{f_b}{4}$$

M-ary Encoding (For 16-QAM, M = 16)

$$N = \log_2 M = \log_2 16 = 4 \text{ bits}$$

Bandwidth Efficiency

Ratio of the transmission bit rate to the minimum bandwidth required for a particular modulation scheme. It is used to compare the performance of one digital modulation technique to the other. It is otherwise known as *information density*.

$$\text{BW efficiency} = \frac{\text{transmission rate (bps)}}{\text{minimum bandwidth (Hz)}}$$

$$\text{BW efficiency} = \frac{f_b}{f_N} \left(\frac{\text{bits}}{\text{cycle}} \right)$$

Carrier Recovery

- Is the process of extracting a phase-coherent reference carrier from a receiver signal.
- Also called phase referencing
- With PSK and QAM, the carrier is suppressed in the balanced modulators and is not transmitted. At the receiver, the carrier cannot be simply tracked with standard PLL.

Three Methods of Carrier Recovery

- Squaring Loop** is a common method of achieving carrier recovery for BPSK.
- Costas Loop** is also called quadrature loop.
- Remodulator** has the fastest acquisition time.

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